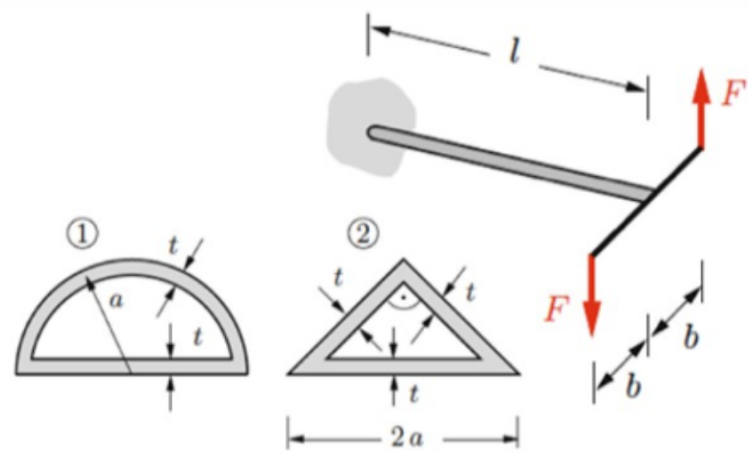


Homework 5.3 The assembly shown in figure consists of a thin-walled elastic tube (shear modulus G) and a rigid lever. The lever is subjected to a couple. Given the allowable shear stress τ_{allow} , calculate the allowable forces and then corresponding angles of twist for the cross sections.



$$\tau = \frac{M_T}{W_T} \quad W_T = 2A_m t \sin \quad \text{since } t \text{ is constant} \quad W_T = 2A_m t$$

$$\tau_{max} = \frac{2bF}{W_T} \quad \Delta\theta = \frac{M_T l}{G I_T} = \frac{2bF l}{G I_T} \quad \Delta\theta_{allow} = \frac{2bF_{allow} l}{G I_T}$$

$$\tau_{max} \leq \tau_{allow} \quad \frac{2bF}{W_T} \leq \tau_{allow} \quad F \leq \frac{W_T \tau_{allow}}{2b} \quad F_{allow} = \frac{W_T \tau_{allow}}{2b}$$

Cross section (1)

$$A_m = \frac{\pi}{2} a^2 \quad \oint \frac{1}{t} ds = \frac{\pi a}{2t} + \frac{2a}{t} = \frac{a}{t} (\pi + 2)$$

$$W_T = \left(\frac{\pi}{2} a^2 \right) t = \pi a^2 t \quad I_T = \frac{(2A_m)^2}{\oint \frac{ds}{t}} = \frac{\pi^2 a^4}{\frac{a}{t} (\pi + 2)} = \frac{\pi^2 a^3 t}{(\pi + 2)}$$

$$F_{allow} = \frac{\pi a^2 t \tau_{allow}}{2b} \quad \Delta\theta_{allow} = \frac{\tau_{allow} (\pi + 2) l}{\pi a G}$$

Cross section (2)

$$A_m = a^2 \quad \oint \frac{1}{t} ds = \frac{2\sqrt{3}a}{t} + \frac{2a}{t} = \frac{a}{t} (2\sqrt{3} + 2)$$



$$I_T = \frac{4a^4}{\frac{a}{t} (2\sqrt{3} + 2)} = \frac{2a^3 t}{\sqrt{3} + 1}$$

$$W_T = 2a^2 t$$

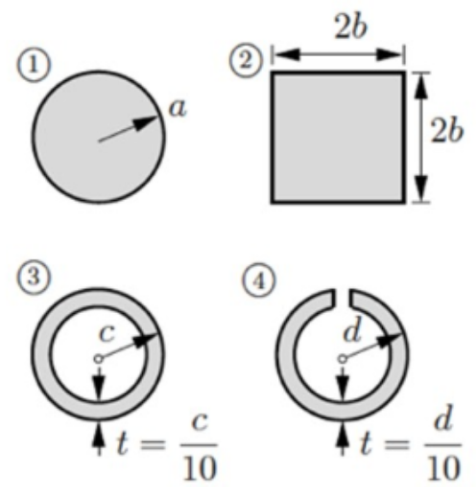
$$F_{allow} = \frac{2a^2 t \tau_{allow}}{2b} = \frac{a^2 t \tau_{allow}}{b}$$

$$\Delta\theta_{allow} = \frac{\tau_{allow} l (\sqrt{3} + 1)}{G a}$$

$$= \frac{(\sqrt{3} + 1) l \tau_{allow}}{G a}$$

Homework 5.4 A shaft has to carry the torque $M_T = 12 \cdot 10^3 \text{ Nm}$. Select a cross section from the depicted group.

Dimension the cross sections such that the admissible shear stress $\tau_{\text{admis}} = 50 \text{ MPa}$ is not exceeded. Which cross section is the most efficient in terms of material usage?



$$\tau_{\max} = \frac{M_T}{W_T} = \tau_{\text{admis}} = 50 \text{ MPa}$$

$$W_{T1} = \frac{J_T}{r} = \frac{\frac{\pi}{2} a^4}{a} = \frac{\pi}{2} a^3$$

$$W_{T2} = \frac{0.141 (2b)^3}{2b} = 0.141 (2b)^3 = 0.141 (8b^3) = 1.128 b^3$$

$$W_{T3} = \frac{\frac{\pi}{2} (c^4 - (c-t)^4)}{c} = \frac{\frac{\pi}{2} (c^4 - (\frac{9}{10}c)^4)}{c} = \frac{\pi}{2} (c^3 - (\frac{9}{10})^4 c^3) = \frac{\pi}{2} (0.3439 c^3) = 0.5402 c^3$$

$$W_{T4} = \frac{2\pi}{3} d t^2 = \frac{\pi}{150} d^3$$

$$a = \sqrt[3]{\frac{2 M_T}{\pi \tau_{\text{admis}}}} = 53.46 \text{ mm}$$

$$b = \sqrt[3]{\frac{M_T}{1.128 \tau_{\text{admis}}}} = 59.7 \text{ mm}$$

$$c = \sqrt[3]{\frac{M_T}{0.5402 \tau_{\text{admis}}}} = 76.3 \text{ mm}$$

$$d = \sqrt[3]{\frac{150 M_T}{\pi \tau_{\text{admis}}}} = 225.45 \text{ mm}$$

$$A_1 = \pi a^2 = 89.7858 \text{ cm}^2$$

$$A_2 = 4b^2 = 142.5656 \text{ cm}^2$$

$$\begin{aligned} A_3 &= \pi (c^2 - (c-t)^2) \\ &= \pi (c^2 - (\frac{9}{10}c)^2) \\ &= \pi (0.19 c^2) \\ &= 34.7498 \text{ cm}^2 \end{aligned}$$

$$A_4 = \frac{\pi}{5} d^2 = 319.36 \text{ cm}^2$$

Third cross section is the most material efficient profile.